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Multi-species time-history measurements during *n*-dodecane oxidation behind reflected shock waves

D.F. Davidson *, Z. Hong, G.L. Pilla, A. Farooq, R.D. Cook, R.K. Hanson

Mechanical Engineering Department, Stanford University, Stanford, CA 94305, United States

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Abstract

Concentration time-histories were measured behind reflected shock waves during *n*-dodecane oxidation for five species: *n*-dodecane, C_2H_4, OH, CO_2, and H_2O. Experiments were conducted at temperatures of 1300–1600 K and pressures near 2 atm using mixtures of 400 ppm *n*-dodecane and 7400 ppm oxygen ($\phi = 1$) in argon. *n*-Dodecane and ethylene were monitored using IR gas laser absorption at 3.39 and 10.53 μm, respectively; OH was monitored using UV laser absorption at 306.5 nm; and CO_2 and H_2O were monitored using tunable IR diode laser absorption at 2.7 and 2.5 μm, respectively. These time-histories provide kinetic targets to test and refine large reaction mechanisms for *n*-dodecane and demonstrate the potential of this type of data for validation of large reaction mechanisms. Comparisons are made with the predictions of two recently developed large-alkane reaction mechanisms, and the need for improved rate measurements of higher alkene reactions is discussed.
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Keywords: Shock tube; Laser absorption; *n*-Dodecane; Reaction mechanisms

1. Introduction

n-Dodecane is used as both a single-component surrogate and a major normal-alkane representative in multi-component surrogates for jet fuel [1]. As such, there is a critical need for reliable kinetic targets to validate and refine reaction mechanisms describing the oxidation of this fuel. Historically, this database of kinetic targets has included shock tube ignition delay times, flow reactor stable species measurements, and species distributions in flames [1–3]. Each of these types of data has advantages, but none gives a complete picture of the internal workings of the chemistry of these processes. Measurements of concentration time-histories for many species, including transient radicals, performed under similar experimental conditions, where chemistry alone dominates the reaction processes, are critically needed. Shock tubes used with laser absorption diagnostics can now provide this type of data.

Laser absorption measurements in shock tubes can provide three major types of kinetic targets: ignition delay times, species concentration time-histories and direct measurements of key elementary reaction rates [4,5]. Studies in our laboratory have previously investigated ignition delay times for *n*-dodecane [3], but there is still only a limited amount of species concentration time-history data (high-pressure OH measurements by Vasu et al. [5]). More complete species measurements would play a critical role in testing and validating large

* Corresponding author.
E-mail address: dfd@stanford.edu (D.F. Davidson).

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n-dodecane mechanisms and refining their component sub-mechanisms.

Operated under optimized conditions, reflected shock tube experiments provide a test environment that can closely approximate a constant volume reactor. Such flows are effectively zero- or one-dimensional and do not contribute significant fluid mechanical, turbulent, or heat transfer effects to the reacting volume. They allow for the isolation of the desired chemical phenomena in a quiescent, high-temperature environment that is very well-characterized. In our laboratory over the past 25 years, we have advanced the study of combustion chemical systems by combining shock tube methods with sensitive laser diagnostic methods.

Laser absorption methods permit measurements of species time-histories that are highly sensitive, species-specific, non-intrusive, and quantitative. Using methods previously developed in our laboratory [6] we are able to probe spectral transitions in the visible and ultraviolet for species such as OH and CH_3 [7,8]. Recent advances in the area of tunable diode lasers have extended these methods to monitor rovibrational transitions in the infrared [9] and allow access to reaction progress markers such as H_2O and CO_2.

Representative computed species time-histories during *n*-dodecane oxidation area are shown in Fig. 1. These time-histories were calculated using the Sirjean et al./JetSurF 1.0 mechanism [10] for several different classes of species: reactants, combustion progress markers or combustion products, small radical species, and stable intermediate products, during reaction under a constant volume and constant internal energy constraint for oxidation of *n*-dodecane at 1450 K, 2.4 atm. Several time scales and reaction intervals are evident. At early times (less than about 10 μs), the fuel decomposition rates and decomposition pathways determine the composition and plateau levels (10–500 μs) of stable intermediates. Here, measurement of the *n*-dodecane and ethylene time-histories places strong constraints on the reaction rates used to describe this process. Following the fuel decomposition, the plateau levels of H-atoms, O-atoms and OH (i.e., at times near 100 μs), are directly related to the rate of reaction progress, e.g. the ignition delay time. Measurements of OH time-histories provide a direct measurement of these small radical pool populations. Nearer the ignition time, higher concentrations of H_2O are observed, indicative of rapid reaction progress. Hence, measurement of both a reaction progress marker, such as H_2O, and a representative small radical pool species, such as OH, places strong constraints on the rate of oxidation processes that are occurring during the ignition induction delay. Constraints on the post-ignition behavior of the system can be derived from the time-history of CO_2, which continues to form at longer times.

It might be assumed that a mechanism for and the successful simulation of *n*-dodecane oxidation could be based on existing mechanisms and their successful simulations of *n*-heptane oxidation [11]. Both fuels are long-chain normal alkanes, and both have been demonstrated to have similar modeled ignition delay time behavior at high temperatures [12]. The major alkene decomposition products of *n*-dodecane are also similar to those seen for *n*-heptane (ethylene, propene, butene). However, in the JetSurF mechanism for *n*-dodecane, the high-temperature decomposition product ratio $[C_3H_6]/[C_2H_4]$ during oxidation is 0.24 (Fig. 1, 1450 K), while for *n*-heptane it is 0.13 (Fig. 2, 1500 K). (This large change in product yields from *n*-dodecane to *n*-heptane is not seen in the LLNL C16 mechanism simulations [12].) Clearly, the decomposition product distribution for *n*-dodecane is strongly coupled to the small radical pool time-histories, and such measure-

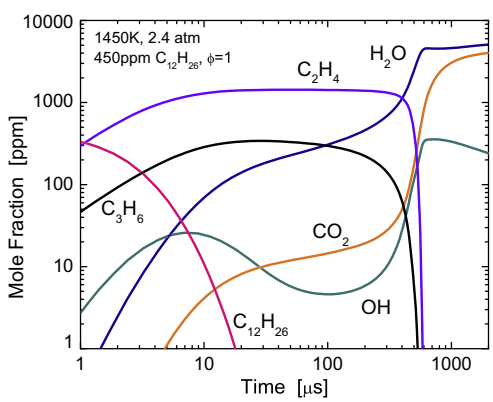

Fig. 1. Calculated species time-histories for *n*-dodecane oxidation using the Sirjean et al./JetSurF 1.0 mechanism. Initial conditions: 1450 K, 2.4 atm, 450 ppm $C_{12}H_{26}/O_2$/argon, $\phi = 1$.

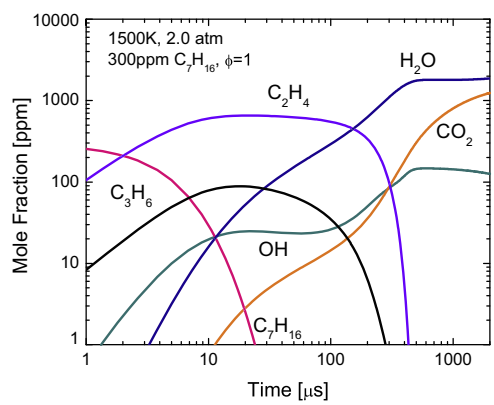

Fig. 2. Calculated species time-histories for *n*-heptane oxidation using the Sirjean et al./JetSurF 1.0 mechanism. Initial conditions: 1500 K, 2.0 atm, 300 ppm C_7H_{16}/O_2/argon, $\phi = 1$.

ments are key to confirming the ability of a mechanism to properly model ignition delay times.

Here we present a high-temperature study of *n*-dodecane oxidation behind reflected shock waves using laser absorption methods to measure time-histories of five species: *n*-dodecane, C_2H_4, OH, CO_2, and H_2O. These measurements are compared to the predictions of two current mechanisms and the important role of alkenes in these mechanisms are discussed.

2. Experimental setup

2.1. Shock tube facility

Experiments were performed in the 6 in. (15.24 cm I.D.) shock tube at Stanford. This shock tube has an electro-polished stainless steel driven section with length/diameter (L/D) ratio of 70 and used a helium-filled driver section with an L/D of 30 to burst 0.25 mm thick polycarbonate diaphragms (against a crossed-knife arrangement in a short square section of the shock tube). The shock tube is sealed with Viton o-rings, is operated at room temperature (i.e., without heating) and can be turbo-pumped overnight to an ultimate pressure $\sim 10^{-6}$ torr and to a leak and outgassing rate of $\sim 3 \times 10^{-6}$ torr/min. Typically reflected shock wave experiments were performed hourly and over this time-frame the ultimate shock tube pressure before filling was $\sim 5 \times 10^{-6}$ torr and the leak and outgassing rate was $\sim 10^{-5}$ torr/min.

Test gas mixtures were stored in a 40 l stainless steel tank heated uniformly to 60 °C, and mixed with a magnetically-driven stirring vane. Mixtures were made using partial pressures and pre-shock filling concentrations in the shock tube were confirmed using *in situ* 3.39 μm laser absorption [13]. 99.999% Argon and 99.99% oxygen (Praxair) were used with spectroscopic grade (99.9+%) *n*-dodecane (Sigma–Aldrich) with no further preparation except for vacuum pumping to removed dissolved volatiles and oxygen.

Incident shock speeds were determined by extrapolating, to the end wall position, shock velocities measured over the last 2 m of the driven section. Reflected shock conditions were calculated using a standard chemically-frozen shock code and standard thermo-chemical data [14]. Resulting uncertainties in reflected shock temperatures over test times of 1 ms were $<\pm 1\%$.

2.2. Laser absorption measurements

Laser absorption of OH: OH concentration was measured using narrow-linewidth laser absorption near 306.7 nm. The chosen wavelength was the peak of the well-characterized $R_1(5)$ absorption line in the OH A-$X(0,0)$ band [8]. CW laser light at 613.4 nm was generated using a Spectra Physics 380 ring dye laser pumped with a 5 W Coherent Verdi at 532 nm. Light at 306.7 nm was generated by intracavity frequency-doubling, using a temperature-tuned AD*A crystal. The wavelength was set using a high-precision wavemeter and confirmed by experiments at off-center wavelengths. Common mode rejection was used to reduce laser intensity noise to <0.1%, resulting in a minimum detection sensitivity of <1 ppm for most conditions in this work. OH concentration was calculated using Beer's law:

$$I/I_0 = \exp(-k_\nu P_{\text{total}} X_{\text{OH}} L) \quad (1)$$

where I and I_0 are the transmitted and incident beam intensities, k_ν is the line-center absorption coefficient at 306.7 nm for OH, P_{total} is the total test gas mixture pressure, X_{OH} is the OH mole fraction, and L is the pathlength, 15.24 cm in the current experiments. The estimated uncertainty in measured X_{OH} is $\sim 3\%$.

Measurements were also performed with the laser tuned off the OH absorption line, and with the laser turned off, to verify that there was no significant interfering absorption or emission.

IR diode laser absorption of CO_2 and H_2O: The recent commercial availability of DFB (distributed feedback) IR diode lasers in the wavelength vicinity of 2.5–2.7 μm has allowed the development of a new CO_2 and H_2O absorption diagnostics [15].

Absorption measurements of CO_2 were made using a fixed-wavelength direct-absorption strategy that provided a detection bandwidth of about 1 MHz. Line-strength, self-broadening and Ar-broadening coefficients were measured previously for the $R(28)$ transition (at 2752.5 nm) used [15], and the ability to make accurate measurements of CO_2 concentration has previously been demonstrated in CO_2–Ar mixtures over a temperature range of 300–1500 K [16]. Quantitative CO_2 concentration profiles were generated from the raw traces of fractional absorption using Beer's law and the known absorption coefficient.

Using the same DFB laser technology at wavelengths of 2.5 μm, the detection of water is also possible [17]. Water concentration time-histories were measured using the IR water transition centered at 2550.96 nm (3920.089 cm^{-1}) within the ν_3 fundamental vibrational band. This line was selected primarily because of its favorable line strength. The line strength and broadening coefficients of this line have been characterized in our laboratory in heated cell experiments before conducting the chemical kinetic studies in a shock tube [9]. In a typical shock tube experiment, with reflected shock temperatures and pressures of 1100 K and 2 atm, 1000 ppm of H_2O in argon over a pathlength of 14 cm gives a strong peak absorbance of approximately 7% with a typical signal-to-noise ratio (SNR) of 50–80.

IR He–Ne laser absorption of n-dodecane: *n*-Dodecane mole fraction was monitored with IR He–Ne laser absorption at 3.39 μm. A C–H

stretch vibrational absorption feature for many hydrocarbons extends across this wavelength. For certain species, large alkanes in particular, absorption at this wavelength is relatively strong; and absorption by smaller hydrocarbon species, and in particular, decomposition products of *n*-dodecane, is weaker. Thus at early times the absorption at 3.39 μm is dominated by the absorption by *n*-dodecane and this measurement can be used to monitor the *n*-dodecane time-history. At later times, after the majority of the *n*-dodecane has decomposed, the residual absorption signal is dominated by the decomposition products with the highest absorption cross-section – mole fraction product. Relatively accurate *n*-dodecane profiles over the entire experimental time-frame are possible, if corrections to the 3.39 μm absorption profiles based on subtraction of ethylene (and higher alkene interfering species) are included. High-temperature absorption cross-sections of *n*-dodecane and its decomposition products, and experimental details of this method, are discussed in Klingbeil et al. [18].

CO_2 laser absorption of C_2H_4: Ethylene mole fraction was monitored by taking advantage of the fortuitous overlap of the *P*(14) line of the CO_2 gas laser at 10.532 μm with the strong ethylene absorption band near 10.6 μm. Similar to the situation with the C–H stretch absorption band near 3.39 μm, there is a wide CH_2 wag absorption feature near 10.6 μm that is shared in varying (but weaker) strengths (at the *P*(14) line in particular) by other alkenes (e.g. ethylene, propene, butene, etc.) However, over the experimental time-frame of the current experiments, absorption at 10.532 μm is dominated by ethylene absorption. High-temperature absorption cross-sections of ethylene and related alkenes (as well as CO_2 and H_2O) and experimental details are discussed in Pilla et al. [19].

3. Results and discussion

Figure 3 shows representative measurements of five measured species time-histories from separate reflected shock wave experiments with initial mixtures of 440 ppm *n*-dodecane/O_2/argon, $\phi = 1$, at nominal temperatures and pressures of 1410 K and 2.35 atm. Small variations in reflected shock conditions were seen over these separate laser absorption experiments. Exact conditions for each experiment are given in the Supplementary data along with species time-histories for intermediate and lower temperature cases. The comparisons between modeled and experimental time-history traces discussed in this paper were based on simulations using the exact conditions under which each species profile was acquired.

Of note is the high signal-to-noise ratio of the measurements and wide range of concentrations and measured time scales possible with these

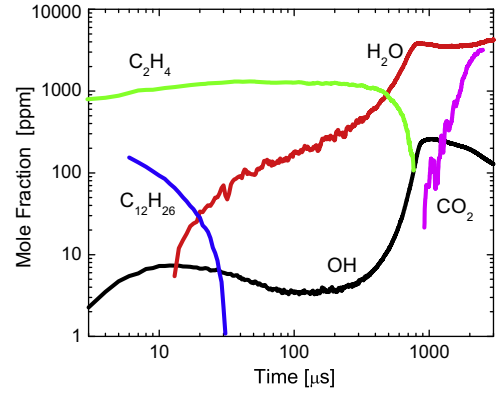

Fig. 3. Species time-history measurements for *n*-dodecane oxidation. Nominal initial conditions: 1410 K, 2.3 atm, 457 ppm *n*-dodecane/O_2/argon, $\phi = 1$.

shock tube/laser absorption techniques. Measurements of OH concentration extend from ppm to 300 ppm levels. Similarly the IR laser diagnostics for *n*-dodecane, ethylene, H_2O and CO_2, while not as sensitive as the OH measurements, but with typically about 50 ppm minimum detectivity, span close to two orders of magnitude in concentration.

As a first application of these species time-history measurements, a comparison of ignition delay times among different species and models was performed.

Figure 4 summarizes the ignition delay times of the full data set and simulations using both the JetSurF 1.0 [10] and LLNL C8-C16 [12] mechanisms. The JetSurF predictions are, in general, longer than the measurements (2×); the LLNL predictions are, in general, shorter (0.5×). Ignition delay times are defined as the time to half-peak concentration for OH, H_2O, and C_2H_4. (Times for *n*-dodecane removal and CO_2 formation are not shown as they occur on significantly different time scales.) These ignition delay times reflect the sequence of removal and formation time-frames of these three species. Neither mechanism studied recovered the measured time-order of formation and removal for these species. In the case of the JetSurF simulation: C_2H_4 is removed first, H_2O, which has been continually forming during the entire ignition process, then achieves its intermediate level, and finally, the OH concentration achieves its peak values. In the case of the LLNL simulation, the order is reversed, though the differences in half-peak values (for the three species) are small in both simulations: ±10% at the lowest temperature and ±20% at the highest temperature. However, the measured ignition delay times for these three species range over a much larger span: a factor of 2.6 at the higher temperatures to 1.6 at the lower temperatures. As well, the activation energies for both models (for all three species) are similar to the measured C_2H_4 ignition

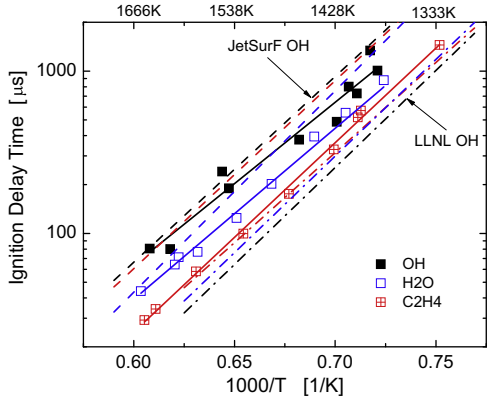

Fig. 4. Ignition delay time measurements for *n*-dodecane, 400 ppm, $\phi = 1$, 2.25 atm. Ignition delay time defined as time to half-peak concentration for OH, H_2O or C_2H_4. Dashed lines: JetSurF simulations; dash-dot lines; LLNL simulations; solid lines; best fit to data. All measurements normalized to the common condition using P^{-1} and ϕ^1 scaling.

delay time activation energy, but differ from the measured OH activation energy.

As a second application of these measurements, the individual species time-histories were compared to model predictions; see Figs. 5 and 6.

This comparison shows that the predicted intermediate C_2H_4 yields and the post-ignition OH, H_2O and CO_2 yields are accurate for both mechanisms. However, except perhaps for the JetSurF CO_2 profile and the general agreement in temporal shape, the current models do not match the time-histories of these four species. The agreement of the post-ignition OH, H_2O and CO_2 peak or plateau values is related to the strong thermo-chemical constraints placed on the quasi-steady-state product concentrations; these values are not strongly sensitive to

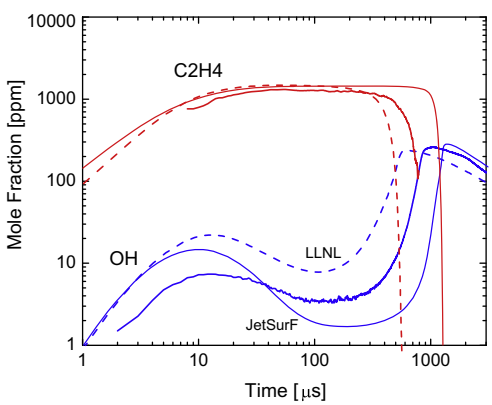

Fig. 5. Species time-histories for OH and C_2H_4. Initial reflected shock conditions: 1410 K, 2.37 atm, 457 ppm *n*-dodecane, 7577 ppm O_2/argon (average conditions of OH, C_2H_4, CO_2, and H_2O experiments). Solid line: JetSurF simulation; dashed line: LLNL simulation.

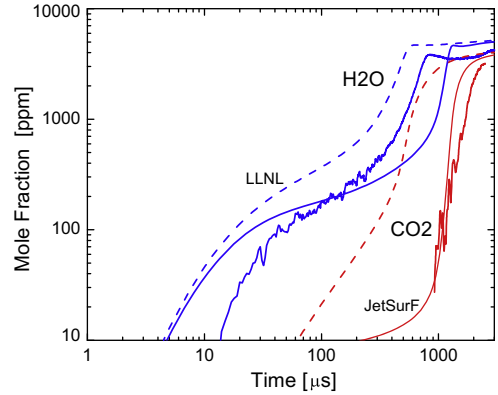

Fig. 6. Species time-histories for H_2O and CO_2. Initial reflected shock conditions: 1410 K, 2.37 atm, 457 ppm *n*-dodecane, 7577 ppm O_2/argon (average conditions of OH, C_2H_4, CO_2, and H_2O experiments). Solid line: JetSurF simulation; dashed line: LLNL simulation.

the reaction mechanism used in the simulations. The agreement of the C_2H_4 yields with the simulations is an indication that the branching ratios of major pathways of the *n*-dodecane decomposition to form C_2H_4 are likely correct. However, the C_2H_4 yield accounts for only ~52% of the carbon carried by intermediates; another 50% of that amount (~26% of the total carbon carried by intermediates) is carried by higher alkenes.

As suggested in Section 1, the higher alkene C_3H_6 (and its decomposition product, the allyl radical aC_3H_5) can play an important role in determining both the ignition delay time and the small radical pool concentrations (represented by OH). A brute force sensitivity analysis (Fig. 7 using the JetSurF mechanism) of the ignition delay time (based on critical reactions which occur in the sensitivity analyses of $C_{12}H_{26}$, C_2H_4, H_2O, CO_2, OH, C_3H_6, CH_3, and CH_4) and a direct sensitivity analysis for OH (Fig. 8) identify the following alkene-related reactions as important (reaction numbering from JetSurf 1.0 [10]):

305. $aC_3H_4 + H = aC_3H_5$
325. $aC_3H_5 + H + M = C_3H_6 + M$
360. $C_3H_6 + H = aC_3H_5 + H_2$
369. $C_3H_6 + OH = aC_3H_5 + H_2O$
252. $C_2H_4 + O = CH_3 + HCO$
254. $C_2H_4 + OH = C_2H_3 + H_2O$
249. $C_2H_4 + H + M = C_2H_5 + M$
194. $C_2H_3 + O_2 = CH_2CHO + O$
1408. $C_{12}H_{26} = nC_3H_7 + pxC_9H_{19}$
1411. $C_{12}H_{26} = 2pxC_6H_{13}$.

Uncertainties in all these reactions are given as either $\Delta \log k = 0.2$ or 0.3 based on the recent review by Baulch et al. [20]. The agreement of

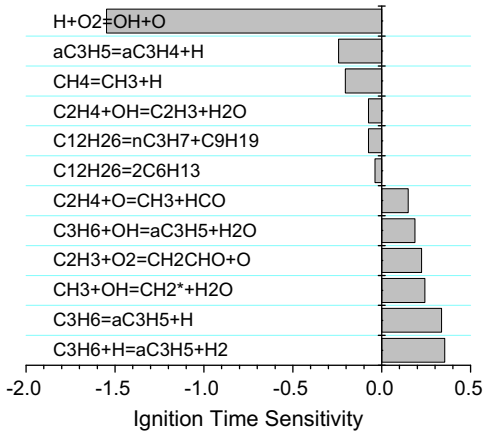

Fig. 7. Ignition delay time sensitivity using the JetSurF 1.0 mechanism. Initial conditions: 1450 K, 2.4 atm, 450 ppm $C_{12}H_{26}/O_2/Ar$, $\phi = 1.0$. Ignition delay time sensitivity is defined as $\{\Delta[\tau_{ign}]/\Delta k_i\}\{k_i/\tau_{ign}\}$, where k_i is the ith reaction rate constant and τ_{ign} is the ignition delay time.

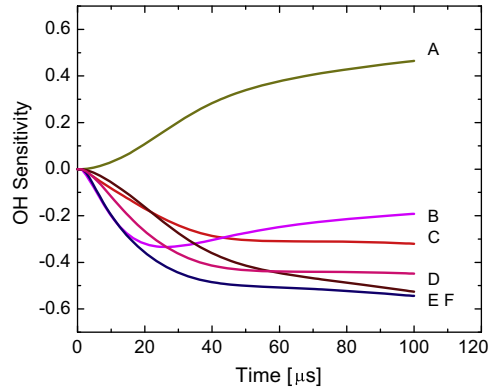

Fig. 8. Details of the early-time sensitivity of OH for the conditions of Fig. 7. The strong sensitivity to the rate of the reaction $H + O_2 = OH + H$ is not shown. Reactions – A (#305): $aC_3H_4 + H = aC_3H_5$; B (#254): $C_2H_4 + OH = C_2H_3 + H_2O$; C: $CH_3 + H + M = CH_4 + M$; D (#369): $C_3H_6 + OH = aC_3H_5 + H_2O$; E (#360): $C_3H_6 + H = aC_3H_5 + H_2$; F (#325): $aC_3H_5 + H + M = C_3H_6 + M$. OH sensitivity is defined as $\{d[OH]/dk_i\}\{k_i/[OH](t)\}$, where k_i is the ith reaction rate constant.

both mechanisms with the C_2H_4 yields, their disagreement with the measured OH profiles, and the generally sparse experiment data available for these particular reactions, points to a strong need for direct study of the rates of these reactions if improved mechanisms for the higher alkanes are to be achieved.

As noted earlier, neither of the two models recovers the wider range of time scales that were measured for C_2H_4, OH, and H_2O. The difficulty that these two mechanisms have matching the time scales for removal of C_2H_4 and formation of H_2O and CO_2 is likely to be directly related to the inability of these mechanisms to duplicate temperature profiles and heat release rates in *n*-dodecane flames [1]. At high temperatures, this inability to match species time scales can translate into uncertainties in the location and width of the reaction zone in *n*-dodecane flames.

4. Conclusion

Measurements of concentration time-histories for *n*-dodecane oxidation were demonstrated for five species: *n*-dodecane, OH, C_2H_4, H_2O, and CO_2. The wide dynamic range of each of these five laser absorption diagnostics in low-concentration experiments allows them to place strong constraints on model predictions over extended time periods. The OH measurements extend from the sub-ppm level to hundreds of ppms, typically over more than two orders of magnitude. Current capabilities with the H_2O diagnostic provides detection capability over 1.5 orders of magnitude (demonstrated here as 70–2000 ppm) with similar detection capabilities for CO_2 possible. Ethylene and fuel concentration measurements are possible from the 100 ppm level to the percent level. These detection ranges are possible because of the very high signal-to-noise ratio measurements achievable with laser absorption techniques. If needed, these detection ranges can be extended to higher concentration by the selection of spectroscopically weaker absorption features.

A comparison of the experimental and modeled species time-histories for *n*-dodecane oxidation shows that while the fuel decomposition pathway reactions in both the JetSurF 1.0 and the LLNL mechanisms allow for an accurate description of ethylene yields, neither mechanism is able to properly model the small radical pool concentrations, in particular the OH species time-histories. The strong sensitivity of the OH time-histories to the H-abstraction and oxidation reaction rates of ethylene and the higher alkenes, and the generally sparse experiment data available for these rates, points to the need for further study of these reactions.

Currently we are extending this multi-species data set to include concentration time-histories for small alkanes (i.e., methane) and other stable hydrocarbon intermediates (i.e., propene and butene) to give a fuller picture of *n*-alkane combustion. Work is also in progress to provide a similar dataset for methylcyclohexane, another important jet fuel surrogate components and to extend all data sets to different temperatures, pressures and stoichiometries.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.proci.2010.05.104.

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